

From a failed certificate to a risk-aware mechanism: A source audit and conditional repair of Q7 through P1

Pathfinder research note

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Abstract

We check the Q7P1 investigation directly against the cached TeX and public version-one HTML of its source papers. Q7’s strict matrix certificate is incompatible with its implementation equalities. Its explicit payment also fails its stated participation condition, and a deterministic example gives an equilibrium with negative participation utility. A separate printed reduction used for learning changes the sign of own-price curvature. These findings do not invalidate the entire mechanism: direct price optimality conditions recover welfare implementation under a local concavity condition. An opponent-only transfer correction then restores equilibrium budget balance and individual rationality without changing best responses. P1 motivates targeting a fixed expected empirical CVaR surrogate, with inward queries that preserve the original allocation set and its zero-allocation opt-out. We give explicit error bounds and a monotone-operator replacement for the failed full-message certificate. This is a source-checked mathematical proposal, not an externally reviewed paper or a validated implementation of Q7’s original stochastic learner.

1 Scope and source identification

Q7 is Garjani, Shokri, and Kebriaei’s mechanism-design paper [1]; P1 is Wang, Huang, Xu, and Johansson’s distributed CVaR paper [2]. Both public records list a single version at the time of this audit. The cached sources are `sources/2608.29130.tex` and `sources/2609.04460.tex`. The relevant displayed expressions agree with the public version-one HTML. The PDF endpoints could not be retrieved through the browsing tool; this is not a claim of a byte-for-byte PDF audit.

The explicit Q7 payment is numbered (24) in the public HTML; its TeX label is `eq40`. References to “equation 40” in earlier working notes should be understood as that source label, not the rendered equation number. The existing Q7P1 follow-up already proposed an opponent-only payment correction and a weighted monotonicity argument. This note rederives those ideas from the sources, sharpens the scope of the negative findings, and adds a direct price-equilibrium argument separating the static repair from the stronger condition needed for the proposed learner. It does not claim that this audit independently originated every ingredient.

We write $\gamma > 0$ for the payment scale, reserving $m > 0$ for valuation curvature. Q7 uses the same symbol α for both. Let $N \geq 2$, let $x_n \in X_n \subset \mathbb{R}_{\geq 0}^K$, and let $p_n \in \mathbb{R}_{\geq 0}^K$. The local sets are convex and compact and contain zero. Prices in the static game are not capped. The capacity constraint is $\sum_n x_n \leq c$. We assume that the planner problem has a KKT multiplier; a suitable Slater condition suffices. Upper bounds on prices require a separate analysis and are not silently included in the theorems below.

2 What is wrong in Q7, as stated

2.1 The matrix conditions have no common solution

Q7's Proposition 1 requires

$$\Psi + \Psi^\top \preceq -\nu I, \quad \nu > 0, \quad (1)$$

where the price-price block in row n , column m of Ψ is $-A_{nm}^n$. Its implementation condition P2(i) requires

$$\sum_m A_{nm}^n = 0 \quad \text{for every } n. \quad (2)$$

These are public equations (5)(ii) and (7)(i), respectively; the cached source locations are lines 200–216 and 240–248.

Proposition 1 (Certificate incompatibility). *No coefficients satisfy both (1) and (2).*

Proof. For nonzero $v \in \mathbb{R}^K$, take $d = \text{col}((0, v), \dots, (0, v))$. Then

$$d^\top \Psi d = - \sum_n v^\top \left(\sum_m A_{nm}^n \right) v = 0. \quad (3)$$

Consequently $d^\top (\Psi + \Psi^\top) d = 0$, contradicting the upper bound $-\nu \|d\|^2 < 0$. $\square$

This is an algebraic contradiction, not a failed numerical search for coefficients. It makes the combined sufficient conditions vacuous and invalidates the assertion that the displayed example satisfies them. It does not prove that the mechanism is unstable. Nor does it establish $\Psi d = 0$: the allocation components can respond to a common-price shift. The obstruction is a zero quadratic form, not necessarily a null vector of the full operator. Replacing strict negativity by a non-strict inequality without analysing the cross terms is not a repair.

2.2 The explicit payment violates participation

Write $S = \sum_m x_m$, $P_{-n} = \sum_{m \neq n} p_m$, and $H_{-n} = \sum_{m \neq n} p_m^\top x_m$. Q7's explicit payment is

$$\begin{aligned} t_n^Q = \gamma & \left[\frac{1}{2} \|p_n\|^2 - \frac{p_n^\top P_{-n}}{N-1} - \frac{p_n^\top (S - c)}{N-1} \right. \\ & \left. + \frac{N P_{-n}^\top (x_n - c/N)}{(N-1)^2} + \frac{\sum_{m \neq n} \|p_m\|^2}{2(N-1)} - \frac{H_{-n} - P_{-n}^\top c/N}{(N-1)^2} \right]. \end{aligned} \quad (4)$$

The TeX source is lines 410–426. Expanding its linear-price terms gives

$$\sum_m a_m^n = \frac{\gamma c}{N(N-1)}. \quad (5)$$

For a positive capacity coordinate this violates Q7's condition P4(ii), which requires that sum to be nonpositive. Thus the explicit formula does not satisfy the advertised sufficient conditions. The following example also shows an actual failure of its claimed participation property.

Proposition 2 (A deterministic participation counterexample). *Set $N = 3$, $K = 1$, $c = \gamma = 1$, $X_n = [0, 1]$, and*

$$V_1(x) = 10x - \frac{1}{2}x^2, \quad V_2(x) = V_3(x) = \frac{1}{10}x - \frac{1}{2}x^2. \quad (6)$$

The profile $x = (1, 0, 0)$, $p = (9, 9, 9)$ is a Nash equilibrium for (4), but agent one's utility is -1 .

Proof. The valuations are smooth, normalised at zero, and 1-strongly concave. The sets are compact and convex, contain zero, and admit strict aggregate slack. Deterministic satisfaction is a permissible special case of Q7’s stochastic model. Each player’s own negative-utility Hessian is

$$\begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}, \quad (7)$$

which is positive definite. At the displayed profile, the price derivatives of negative utility are zero and its allocation derivatives are $(0, 89/10, 89/10)$. These satisfy the box-constrained first-order conditions, which are globally sufficient by convexity. Direct substitution gives

$$t^Q = (21/2, -21/4, -21/4), \quad V = (19/2, 0, 0), \quad (8)$$

so the first utility is $19/2 - 21/2 = -1$. $\square$

The example does not satisfy P4(ii), so it is not a counterexample to a conditional participation argument whose premises include P4(ii) and the required equilibrium structure. It refutes the explicit formula’s claimed participation guarantee and the assertion that this formula satisfies P4. This distinction corrects the broader informal description of the failure.

The same allocation with any common price $p_n = \lambda$, $\lambda \in [1/10, 9]$, satisfies the same sufficient first-order conditions. Thus this instance also directly refutes uniqueness of the full message profile. The allocation is unique; the supporting price need not be.

2.3 The printed learning reduction is not equivalent

In Section 4, Q7 replaces t_n^Q by a reduced payment $\hat{t}_n = \gamma[s_n^\top M s_n + (\Delta \bar{s}_{-n} + d)^\top s_n]$, declaring that only terms independent of the player’s own message have been removed. But its printed M has price-price block $-I/2$ (cached lines 432–440). Therefore

$$\nabla_{p_n p_n}^2 t_n^Q = \gamma I, \quad \nabla_{p_n p_n}^2 \hat{t}_n = -\gamma I. \quad (9)$$

Deleting opponent-only terms cannot change this curvature. This establishes a printed inconsistency, not a finding about unpublished code or author intent. In particular, repairing the static transfers does not establish the stated stochastic-learning theorem. The replacement below uses the derivatives of the explicit payment, not this reduced expression.

3 A static repair that preserves incentives

Define the opponent-only correction

$$r_n(s_{-n}) = \frac{\gamma}{(N-1)^2} \left(P_{-n}^\top c - N H_{-n} \right), \quad t_n^R = t_n^Q - r_n(s_{-n}). \quad (10)$$

The correction can have either sign. Because it is independent of (x_n, p_n) , it leaves every unilateral payoff difference, best response, and Nash equilibrium unchanged for a fixed collection of valuations. A useful exact identity is

$$p_1 = \cdots = p_N = p \implies t_n^R = \gamma p^\top (x_n - c/N), \quad (11)$$

even away from aggregate feasibility.

Theorem 1 (Allocation implementation, balance, and participation). *Suppose each V_n is continuously differentiable and m -strongly concave, with $V_n(0) = 0$, and*

$$m \geq \frac{\gamma}{(N-1)^2}. \quad (12)$$

Under the set and KKT assumptions in Section 1, the game with payments t_n^R has a Nash equilibrium. Every equilibrium implements the unique welfare-maximising allocation, uses a common nonnegative price, balances the budget, and gives each agent utility at least zero. Full message uniqueness is not asserted.

Proof. Let $h = S - c$. At any equilibrium the price first-order conditions, coordinate by coordinate, are

$$p_n \geq 0, \quad g_n := Np_n - \sum_j p_j - h \geq 0, \quad p_n g_n = 0. \quad (13)$$

They follow from the strictly convex own-price part of either payment. At an agent with minimum price, $Np_n - \sum_j p_j \leq 0$; hence $g_n \geq 0$ forces $h \leq 0$. If the maximum price is positive, its complementarity condition yields $0 = Np_{\max} - \sum_j p_j - h$. Both terms on the right are non-negative, so $h = 0$ and all prices equal the maximum. Otherwise all prices are zero. Applied to each resource, this proves common pricing, feasibility, and $p^\top(S - c) = 0$ at every equilibrium.

The own-allocation derivative of the payment is

$$\nabla_{x_n} t_n^R = \gamma \left(-\frac{p_n}{N-1} + \frac{NP_{-n}}{(N-1)^2} \right). \quad (14)$$

At common prices this is γp . The allocation conditions are therefore exactly the planner KKT conditions with multiplier $\lambda = \gamma p$. Strict concavity makes the planner's allocation unique.

Conversely, take a planner optimum and a KKT multiplier λ and set $p_n = \lambda/\gamma$. The own negative utility, up to affine and opponent-only terms, can be written as

$$-V_n(x_n) - \frac{\gamma}{2(N-1)^2} \|x_n\|^2 + \frac{\gamma}{2} \left\| p_n - \frac{x_n}{N-1} \right\|^2. \quad (15)$$

It is convex under (12). The planner KKT conditions thus give globally optimal unilateral choices, proving equilibrium existence.

Finally, (11) and complementarity give $\sum_n t_n^R = \gamma p^\top(S - c) = 0$. An agent can deviate to $(0, p)$ while the other messages remain fixed. This is locally feasible and gives utility $\gamma p^\top c/N \geq 0$. Its equilibrium utility is no smaller. In the counterexample, the corrected taxes are $(6, -3, -3)$ and utilities are $(7/2, 3, 3)$. $\square$

The price argument also applies to Q7's original payment. Under the same concavity and KKT premises, its equilibrium allocations can therefore be rescued without its impossible certificate. The transfer correction repairs participation, not allocation incentives. Budget balance is an equilibrium property; neither payment guarantees physically feasible requests at every off-equilibrium message profile. The proof also explains why unique allocation is the appropriate general conclusion, rather than unique prices.

4 How P1 induces a different mechanism-design target

P1's fixed-batch analysis identifies a deterministic object: the expected empirical, smoothed CVaR, rather than the population CVaR itself. Its one-point estimator is unbiased for the gradient of that surrogate (Section 4.3, cached lines 853–878). This is the useful connexion: design the strategic mechanism for the objective its oracle actually estimates, and then control the difference from the scientific target.

There are genuine interface changes. P1 concerns cooperative agents sharing a decision vector; Q7 has selfish agents with separate resource demands. P1 assumes a ball about zero inside its decision set, whereas a nonnegative resource set generally has zero on its boundary. Translating or shrinking the strategic allocation set would change the meaning of zero consumption and the feasible welfare benchmark. We instead move the *queries*, not the allocations.

4.1 A fixed surrogate with feasible queries

For agent n , let

$$C_n(x) = \text{CVaR}_{\beta_n}[J_n(x, \xi)], \quad 0 < \beta_n < 1. \quad (16)$$

Assume samplewise convexity, L_n -Lipschitz continuity, and $|J_n(x, \xi)| \leq U_n$ on X_n . These are P1-type oracle assumptions, now applied to separate local resource variables. Assume a known interior ball $a_n + \rho_n \mathbb{B} \subset X_n$. Choose $0 < \varepsilon_n < 1$ and $0 < \delta_n < \varepsilon_n \rho_n$ and define

$$q_n(x, u) = (1 - \varepsilon_n)x + \varepsilon_n a_n + \delta_n u. \quad (17)$$

Every query with $\|u\| \leq 1$ is in X_n by convexity. In particular, the message $x = 0$ remains feasible, without querying outside the resource domain. Lower-dimensional sets require a relative-interior construction; the present statement assumes a full-dimensional ball.

Let $\hat{C}_{n,s_n}$ denote empirical CVaR from a fresh i.i.d. batch of fixed size s_n . Freeze s_n , its distribution, and the query parameters. Define

$$\tilde{C}_n(x) = \mathbb{E}_{\nu, \xi^{1:s_n}} \hat{C}_{n,s_n}(q_n(x, \nu)) + \frac{\kappa_n}{2} \|x\|^2, \quad \nu \sim \text{Unif}(\mathbb{B}), \quad (18)$$

and $\tilde{V}_n(x) = \tilde{C}_n(0) - \tilde{C}_n(x)$. The optional $\kappa_n \geq 0$ supplies missing curvature. If every sample loss is already μ_n -strongly convex, the surrogate is strongly convex with modulus at least $\mu_n(1 - \varepsilon_n)^2 + \kappa_n$; for merely convex losses take $\mu_n = 0$. This follows because empirical CVaR is a maximum of nonnegative weighted sample losses whose weights sum to one. Smoothing and expectation preserve the modulus after the affine contraction.

Select γ so that the minimum surrogate modulus satisfies (12). Theorem 1 now applies exactly to the game with valuations $\tilde{V}_n$. No claim that P1's cooperative network algorithm learns this strategic equilibrium is needed or made.

4.2 What is lost relative to true risk

Put $R_n = \sup_{x \in X_n} \|x - a_n\|$, $D_n = \sup_{x \in X_n} \|x\|$, and

$$e_n = \frac{U_n}{\beta_n} \sqrt{\frac{2\pi}{s_n}}. \quad (19)$$

The empirical minimum has downward bias: $\mathbb{E} \hat{C}_{n,s_n}(y) \leq C_n(y)$. Its expected absolute error is at most e_n , by the bounded-loss empirical-CDF estimate used in P1. Consequently, with $b_n(x) = \tilde{C}_n(x) - C_n(x)$,

$$-e_n - L_n \varepsilon_n R_n \leq b_n(x) \leq L_n \varepsilon_n R_n + L_n \delta_n + \frac{\kappa_n D_n^2}{2}. \quad (20)$$

For the lower bound, use Jensen's inequality for ball smoothing about $(1 - \varepsilon_n)x + \varepsilon_n a_n$, then the Lipschitz displacement bound. For the upper bound use downward empirical bias and Lipschitz continuity. Define the interval width

$$w_n = e_n + 2L_n \varepsilon_n R_n + L_n \delta_n + \frac{\kappa_n D_n^2}{2}. \quad (21)$$

Corollary 1 (True-risk consequences). *Let $\tilde{s} = (\tilde{x}, \tilde{p})$ be a surrogate-game equilibrium and let x^* minimise $\sum_n C_n(x_n)$ on the same original resource-feasible set. Then*

$$\sum_n C_n(\tilde{x}_n) - \sum_n C_n(x_n^*) \leq \sum_n w_n, \quad (22)$$

$$C_n(0) - C_n(\tilde{x}_n) - t_n^R(\tilde{s}) \geq -w_n. \quad (23)$$

Moreover, no unilateral deviation in the true-risk game can improve agent n 's utility by more than w_n at $\tilde{s}$.

Proof. Surrogate optimality bounds the true aggregate gap by $\sum_n [b_n(x_n^*) - b_n(\tilde{x}_n)]$, at most $\sum_n w_n$. The difference between true and surrogate normalised valuations is $b_n(x) - b_n(0)$, whose magnitude is at most w_n . For a unilateral deviation, the normalisation constant cancels: the change in this difference is $b_n(x'_n) - b_n(\tilde{x}_n) \leq w_n$. The surrogate deviation gain is nonpositive and the tax is identical in both games. $\square$

Thus equilibrium balance and implementation are exact for the declared surrogate game; participation and incentives relative to true CVaR have explicit errors. If agents actually optimise true CVaR rather than the declared surrogate, $\tilde{s}$ is an approximate equilibrium, not automatically an exact one. Regularisation is also not free: its distortion is included in w_n .

5 Replacing, rather than assuming, the convergence certificate

Let F be the negative-utility pseudogradient for the surrogate game, and group coordinates as (x, p) . Deriving directly from the payments gives

$$F_{x_n} = \nabla \tilde{C}_n(x_n) + \gamma \left(-\frac{p_n}{N-1} + \frac{NP_{-n}}{(N-1)^2} \right), \quad (24)$$

$$F_{p_n} = \frac{\gamma}{N-1} \left(Np_n - \sum_j p_j - S + c \right). \quad (25)$$

Let $D = \text{diag}(I, (N-1)I/N)$ and $G = DF$. Positive scaling of the separate allocation and price blocks preserves their product-set normal-cone conditions, so the zeros of $G + N_{\mathcal{S}}$ are the same Nash equilibria. Write $\Pi = (I_N - \mathbf{1}\mathbf{1}^T/N) \otimes I_K$ and $a = \gamma(2N-1)/(N-1)^2$.

Proposition 3 (A feasible monotonicity replacement). *If the surrogate costs are m -strongly convex and*

$$m > \frac{\gamma(2N-1)^2}{4(N-1)^4}, \quad (26)$$

then G is monotone. In particular, for differences $\Delta x, \Delta p$,

$$\langle G(s) - G(s'), s - s' \rangle \geq m\|\Delta x\|^2 + \gamma\|\Pi\Delta p\|^2 - a\langle \Pi\Delta x, \Pi\Delta p \rangle \quad (27)$$

$$\geq \left(m - \frac{a^2}{4\gamma} \right) \|\Delta x\|^2 + \gamma \left\| \Pi\Delta p - \frac{a}{2\gamma} \Pi\Delta x \right\|^2. \quad (28)$$

Proof. The allocation-price block of the unscaled operator has eigenvalue γ on common prices and $-a$ on price disagreements. The scaled price-allocation block is $-\gamma(\mathbf{1}\mathbf{1}^T/N) \otimes I_K$, so their common-mode contributions to the quadratic form cancel. The scaled price-price block is $\gamma\Pi$. Strong convexity supplies the allocation term, and completing the square gives the second inequality. The positive allocation coefficient follows from (26). $\square$

This deliberately permits a zero common-price quadratic form. It does not contradict the obstruction, and is stronger than the local concavity condition needed for the static theorem. A convergent replacement is the proximal point iteration

$$s^{k+1} = (I + h\mathcal{A})^{-1}s^k, \quad \mathcal{A} = G + N_{\mathcal{S}}, \quad h > 0. \quad (29)$$

Here $N_{\mathcal{S}}$ denotes the convex normal cone. For completeness, $\mathcal{A}$ is maximal monotone: subtract $m\|x_n\|^2/2$ from each surrogate cost and include the set indicators in the resulting proper, closed convex function. Its subdifferential is maximal monotone. The remaining affine operator is monotone by the calculation above and is continuous on the full space, so their sum is maximal monotone. Theorem 1 supplies a zero. The resolvent is firmly nonexpansive; its iterates are Fejer

monotone with respect to the zero set, have vanishing steps, and converge in finite dimensions to a zero [3]. Convergence to one supporting price does not imply that all initialisations select the same price.

An inexact version retains convergence if its errors relative to the exact resolvent are summable. This is a contract for a replacement learner, not a validation of Q7’s parallel proximal best-response iteration: the resolvent solves a coupled variational inequality.

P1 supplies an oracle for the allocation part. With u uniform on the unit sphere in dimension K and a fresh batch, set

$$\hat{g}_n(x) = (1 - \varepsilon_n) \frac{K}{\delta_n} \hat{C}_{n,s_n}(q_n(x, u))u + \kappa_n x. \quad (30)$$

It is conditionally unbiased for $\nabla \tilde{C}_n(x)$, and $\|\hat{g}_n(x)\| \leq (1 - \varepsilon_n)KU_n/\delta_n + \kappa_n D_n$. This supplies a bounded stochastic oracle for inner solves. A practical decentralised inner solver, its sample complexity, and a proof that its errors meet the summability requirement are not supplied here. Changing batches between calls is compatible with the fixed surrogate; changing their size or distribution changes the target and needs another argument.

6 Verification, novelty, and the connexion

The accompanying `check_q7.py` uses rational arithmetic for the counterexample, payment identities, gradients, price continuum examples, and finite-dimensional instances of the monotonicity identity. It also enumerates a two-outcome, two-sample CVaR oracle to check its expected surrogate and inward-query gradient identity. These are reproducible algebra checks, not measurements on Q7’s transportation data or P1’s sensor network, and they do not replace the general proofs.

The source identities, counterexample, and direct equilibrium argument are checked here without assuming the earlier internal review’s verdict. This is still a single audit, not external peer review. The only new literature search in this audit was a targeted check for a Q7 correction and the standard proximal-point reference; no correction was located in that bounded search. The payment invariance principle, convex perturbation estimates, and proximal-point convergence are established ideas. Publication novelty of this particular synthesis has not been established.

The proposed connexion can therefore be stated precisely: *P1’s sampling semantics induce a reformulation of Q7 around a fixed risk surrogate, allocation rather than message uniqueness, and explicitly bounded economic error.* The audit removes a false certificate and repairs the transfer normalisation; the resulting mechanism theorem does not need that certificate. What remains open is the efficient stochastic implementation of the replacement learner and independent validation of the synthesis.

A Audit provenance

The SHA-256 hashes of the cached sources at the read used for this audit are:

Q7: 2a4e5c7f74fb0350d271636307f402af3
b0f734341f798312ae7a6ece9bbbe76
P1: c5631017064059edf3bb128b0bfc9177
3601627142112e48a39dd88f6ba0d803

Each hash is split across consecutive lines for typesetting; concatenate the two hexadecimal fragments without whitespace. Existing campaign manuscripts, review records, and status files were not changed by this audit.

References

- [1] S. Garjani, M. Shokri, and H. Kebriaei. *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*. arXiv:2608.29130v1, 2026. <https://arxiv.org/html/2608.29130v1>.
- [2] S. Wang, K. Huang, L. Xu, and K. H. Johansson. *Distributed risk-averse optimization via CVaR*. arXiv:2609.04460v1, 2026. <https://arxiv.org/html/2609.04460v1>.
- [3] R. T. Rockafellar. Monotone operators and the proximal point algorithm. *SIAM Journal on Control and Optimization*, 14(5):877–898, 1976. <https://doi.org/10.1137/0314056>.